

KeraNova Data Cleaning Rules

Version 1.0 for committee review on 1 October 2026

KeraNova Research Team | Version 1.0 | 1 October 2026

Purpose and current audit

Apply these rules to a derivative dataset while preserving the original CSV, workbook and authorized source exports. Cleaning resolves formatting and records uncertainty. It does not create missing clinical observations, verify an undocumented procedure, or turn a source-labelled visit into a dated endpoint.

The current CSV audit found 79 eye records after removing four administrative headings from analysis, two ages below 18, no exact duplicate full rows, and two MRSE consistency review candidates. It contains only nine baseline and two source-labelled twelve-month CDVA entries. Patient codes and dates are absent, so the primary model dataset cannot yet be released.

Storage and change log

For each clinical variable retain <field>_raw, <field>_clean, <field>_flag and <field>_code. Numeric clean fields contain numeric values or null; missing codes live in the separate code field. The flag field records all applicable rule IDs and review status. Preserve the distinction between a documented value and a calculated derivative.

The change log contains source hash and row, record_id, coded patient_id when recovered, eye_id, variable, rule ID and version, old and new value, reason, extractor or script version, timestamp, reviewer and resolution. Raw files are read only. Save cohort, dictionary, clean data and code versions with each release.

Missing and negative value codes

Code	Meaning	Rule
OBS	Observed usable value	0 and No remain valid observed values
ND	Not documented or blank placeholder	Blank, a standalone hyphen, dot, NA-like source placeholder; retain original phrase
NA	Not applicable	Only when nonapplicability is explicit; source n/a without context requires review
UNK	Explicitly unknown	A source states unknown; distinct from absent information
NR	Source not retrievable	Requested record unavailable; state retrieval attempt
AMBIG	Unresolved meaning or source conflict	Qualified VA or unidentified metric; do not guess
INVALID	Confirmed unusable measurement	Clinical reviewer confirms erroneous or quality-failed value; raw retained

Import identity and dates

Rule	Area	Action
R01	Import	Use UTF-8 with BOM tolerance. Validate 42 unique exact CSV headers and consistent row widths. Preserve original punctuation, whitespace and source hash.
R02	Administrative rows	Exclude source rows 2, 21, 40 and 63 from eye counts only because they have an administrative label and no clinical values. Keep an import disposition log.
R03	Identifiers	Assign source row IDs such as CSV_R0006. They are record IDs, not patient IDs. Recover patient linkage from the authorized custodian; do not infer identity from age, sex or similar scans.
R04	Laterality and text	Trim outer spaces in clean text and normalize Right to OD and Left to OS. Femal to Female is a documented spelling correction. Log every changed value; preserve unknown categories.
R05	Age	Remove y or years suffix and parse a nonnegative age. CSV age timing is unknown. Verify age at treatment and flag values below 18 for the adult-cohort eligibility decision.
R06	Dates	Normalize verified dates to YYYY-MM-DD with source calendar and day-month-year convention recorded. Resolve ambiguity using source evidence; never infer exact dates from 1, 6 or 12 month labels.
R07	Chronology	Index date must fall within approved treatment dates; baseline precedes index, or same-day timing proves preoperative status; follow-up is after index. Flag invalid sequences and values after the authorized cutoff.
R08	Duplicates	Flag exact repeated rows and repeated patient-eye-index keys when IDs are available. Compare source encounter IDs and timestamps; adjudicate conflicting duplicates. Similar age and sex are insufficient to remove a record.

Numeric measurements and clinical context

Rule	Area	Action
R09	Numeric parsing	Accept a dot or a verified decimal comma; distinguish decimal from thousands separators. Strip D or micrometre suffix only in the relevant field. Preserve signed values, zero and original precision.
R10	Refraction signs	A trailing plus such as 0.25+ becomes +0.25 only in a numeric refraction field. PL or Pl in a sphere field is plano under the verified source convention and may become 0 D; it is not light perception.
R11	MRSE	Calculate sphere plus cylinder divided by 2 from same-encounter valid values. Keep reported MRSE separately; difference greater than 0.02 D flags review. Do not overwrite a reported discrepancy automatically.
R12	Visual acuity role	Keep UDVA, CDVA, pinhole and contact-lens acuity distinct. Confirm correction and chart method. Do not replace missing CDVA with a different acuity category or postoperative value.

Rule	Area	Action
R13	Visual acuity conversion	For a verified unqualified Snellen fraction n/d, logMAR = $\log_{10}(d/n)$; for decimal acuity $v > 0$, logMAR = $-\log_{10}(v)$; retain directly documented logMAR after checking its scale. Record conversion method.
R14	Modifiers and low vision	Preserve + or - letter modifiers. Do not assume a per-letter correction without chart/scoring evidence. Unresolved qualified values stay AMBIG for primary analysis. CF, HM, LP and NLP remain categorical; no arbitrary numerical primary label.
R15	Unusual fractions	Fractions such as 4/200 and 25/25-1, and a line labelled 20/24, require chart-distance or transcription verification. Never silently rewrite numerator or denominator to a familiar line.
R16	Topography timing	Postop K1, K2, Kmax, thinnest and PAD-d lack a time interval. Preserve as unassigned postoperative values until dates and encounters are recovered. Do not use them as baseline predictors.
R17	Metric identity	Confirm that thinnest loca denotes TCT in micrometres and that PAD-d denotes BAD-D before the respective mappings. Keep central, apical and thinnest thickness separate.
R18	Scan quality	Use documented scan-quality indicators, laterality, device and scan date. Confirm inconsistent K1/K2 ordering or extreme Kmax against the original scan; retain valid severe-disease measurements.
R19	Treatment	Normalize source labels provisionally. Verify topography guidance and accelerated CXL before final cohort assignment. The PTK record is a different procedure; retain it in screened records with exclusion reason.
R20	Haze and safety	No and No haze are explicit negatives for the recorded observation. Positive haze phrases are recorded as positive text with date and grade unverified. Do not infer resolution or standard grade from Mild, +1 or ++ without a documented scale; adjudicate old haze.
R21	Severity	Accept a documented Amsler Krumeich stage 1 to 4 after verification. Do not reconstruct stage or a mild/severe group from one isolated measurement.

Range review thresholds

The following are conservative data-review triggers, not clinical exclusion criteria. A value outside a trigger range remains in the raw data and requires source review; a clinically verified extreme may remain in the clean dataset. Eligibility thickness limits come from the approved clinical protocol, not this table.

Field	Review trigger	Decision
Age	Less than 0 or greater than 100 years	Under 18 separately triggers adult eligibility review
Sphere or cylinder	Absolute value greater than 20 D	Check sign, unit, transcription and cylinder convention
K1 K2 or Kmax	Less than 20 D or greater than 100 D	Check device and severe disease; Kmax 96.8 D is not an automatic deletion

Field	Review trigger	Decision
Corneal thickness	Less than 100 or greater than 1000 micrometres	Check units and whether value is thickness or location
logMAR	Less than -0.5 or greater than 2.0	Check chart range and correction; do not cap values
Amsler Krumeich stage	Outside 1 to 4	Review documented scale and source

Specific findings requiring source review

Record or issue	Evidence	Required action
CSV row 23 baseline MRSE	Sphere +1.00 D and cylinder -5.00 D yield -1.50 D; reported +1.50 D	Check source for sign error; retain both until adjudicated
CSV row 57 one-month MRSE	Calculated -4.625 D; reported -4.652 D	Difference 0.027 D exceeds 0.02 D trigger; check rounding or transcription
CSV row 6 twelve-month CDVA	20/30-	Resolve modifier, correction and actual visit date
CSV row 76 twelve-month CDVA	20/25	Verify actual date, correction method and patient linkage
CSV rows 20 and 72 ages	17 y and 16y	Verify index age; exclude adult-cohort analysis if confirmed under 18
Similar records such as rows 28 and 49	Matching age, sex and several scans with differing refraction and thickness	Possible duplicate candidate only; no deletion without source identity

Manual review and release criteria

A second reviewer verifies all final treatment assignments, core eligibility criteria and primary outcome labels. Resolve issues using dated records, document the decision and retain a reason when a value becomes missing or a record is excluded. If a source cannot resolve a conflict, retain AMBIG or NR and report the limitation.

Before release, reconcile 83 imported non-header CSV rows to 79 eye records plus four administrative rows, inspect missingness by treatment and interval, verify patient grouping, dates, endpoint selection and identifiers, and check a sample of every transformation rule independently. No global outcome imputation or carry-forward is allowed. Any predictor imputation or scaling for a future model is learned only from its training data within each resample.

Measurement reference

Kaiser PK. Prospective evaluation of visual acuity assessment. Transactions of the American Ophthalmological Society. 2009;107:311–324. <https://pmc.ncbi.nlm.nih.gov/articles/PMC2814576/>. The study supports the reciprocal Snellen logMAR conversion and shows that mathematical conversion does not remove chart-method differences. The conservative handling of unresolved modifiers here is a prespecified KeraNova data-quality rule.